

Disproof of conjecture 00000002048

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Verdict: FALSE

We disprove conjecture 00000002048 as stated. The refutation is unconditional and elementary: the conjecture's main claim already fails at the smallest non-trivial prime, $p = 5$, for the two-element set $A = \{0, 1\}$. A general family shows that the failure is not an isolated accident.

1 The conjecture

Definition 1 (3-fold sum). For a prime p and a set $A \subseteq \mathbb{F}_p$, the *3-fold sum* of A is

$$3A = \{x + y + z \bmod p : x, y, z \in A\} \subseteq \mathbb{F}_p.$$

Quoted verbatim from `conjectures/00000002048.md`:

Definition: The 3-fold sum of $\mathbb{F}_p$ is the coverage of $\{x + y + z : x, y, z \in A\}$. **Conjecture:** For $|A| > (p - 1)/3$, the 3-fold sum covers $\mathbb{F}_p \setminus \{0\}$ (the threshold is exact); the threshold is optimal (attained critically by the cubic-residue set at the threshold).

The claim consists of two parts: a *coverage claim* ($3A \supseteq \mathbb{F}_p \setminus \{0\}$ whenever $|A| > (p - 1)/3$) and an *optimality claim* (the constant $(p - 1)/3$ is exact, with the cubic-residue set attaining it critically). Both fail; the coverage claim fails first.

2 The counterexample at $p = 5$

Theorem 2. Let $p = 5$ and $A = \{0, 1\} \subseteq \mathbb{F}_5$. Then

$$|A| = 2 > \frac{4}{3} = \frac{p-1}{3}, \quad 3A = \{0, 1, 2, 3\}, \quad 4 \notin 3A.$$

In particular $3A$ does not contain the nonzero element 4 of $\mathbb{F}_5$, so $3A \not\supseteq \mathbb{F}_5 \setminus \{0\}$.

Proof. The set A has two elements, and $2 > 4/3$ since $3 \cdot 2 = 6 > 4 = p - 1$. The 3-fold sum is computed by enumerating the $2^3 = 8$ triples $(x, y, z) \in \{0, 1\}^3$: the sum $x + y + z$ ranges over $\{0, 1, 2, 3\}$. No wraparound occurs because $3 < 5$, so

$$3A = \{0 + 0 + 0, 0 + 0 + 1, 0 + 1 + 1, 1 + 1 + 1\} = \{0, 1, 2, 3\}$$

as a subset of $\mathbb{F}_5$. The residue 4 is not in this set: it is not the image of any of the eight triples. Since $4 \neq 0$ in $\mathbb{F}_5$, the element 4 lies in $\mathbb{F}_5 \setminus \{0\}$ but not in $3A$. $\square$

Corollary 3. *Conjecture 00000002048 is false. At $p = 5$ the hypothesis $|A| > (p - 1)/3$ holds strictly for $A = \{0, 1\}$, yet the conclusion fails.*

Remark 4. The threshold is far from exact. At $p = 5$ the hypothesis already holds for $|A| = 2$, and the conclusion fails; a set of size exactly $(p - 1)/3 = 4/3$ does not even exist. Nothing here depends on the structure of A beyond its size and its image under sums.

3 The general family $p \equiv 2 \pmod{3}$

The witness is the first member of an infinite family.

Theorem 5. *Let p be a prime with $p \equiv 2 \pmod{3}$, and let*

$$A = \left\{ 0, 1, \dots, \frac{p-2}{3} \right\} \subseteq \mathbb{F}_p.$$

Then

$$|A| = \frac{p+1}{3} > \frac{p-1}{3}, \quad 3A = \{0, 1, \dots, p-2\}, \quad p-1 \notin 3A.$$

Consequently the conjecture's coverage claim fails for every such prime.

Proof. Write $p = 3k + 2$ with $k \geq 0$, so that $(p - 2)/3 = k$ and $A = \{0, \dots, k\}$. Then $|A| = k + 1 = (p + 1)/3$, and

$$|A| = \frac{p+1}{3} > \frac{p-1}{3}$$

because $p + 1 > p - 1$. This is the hypothesis of the conjecture.

For the sum: every $x \in A$ satisfies $0 \leq x \leq k$, hence every triple $x, y, z \in A$ satisfies

$$0 \leq x + y + z \leq 3k = p - 2 < p.$$

As all sums lie strictly below p , no reduction modulo p occurs, and

$$3A = \{x + y + z : x, y, z \in \{0, \dots, k\}\}.$$

Every integer between 0 and $3k = p - 2$ is a sum of three elements of $\{0, \dots, k\}$: for $t \in \{0, \dots, p - 2\}$ take $a = \min(t, k)$, $b = \min(t - a, k)$ and $c = t - a - b$, so that $a, b, c \in \{0, \dots, k\}$ and $a + b + c = t$ (if $t \leq k$ then $a = t$ and $b = c = 0$; if $t > k$ then $a = k$ and the remainder $t - k \leq 2k$ is split between b and c , each at most k). Hence

$$3A = \{0, 1, \dots, p - 2\}.$$

In particular $p - 1 \notin 3A$. Since $p - 1 \neq 0$ in $\mathbb{F}_p$ (as $p \geq 2$), the element $p - 1 \in \mathbb{F}_p \setminus \{0\}$ is not covered. So $3A \not\supseteq \mathbb{F}_p \setminus \{0\}$. $\square$

Remark 6. For $p = 5$ the family gives $k = 1$, $A = \{0, 1\}$: exactly the witness of Theorem 2, with $3A = \{0, 1, 2, 3\}$ and $p - 1 = 4$ missing.

Corollary 7. *The coverage claim of conjecture 00000002048 is false for every prime $p \equiv 2 \pmod{3}$, in particular for*

$$p = 5, 11, 17, 23, 29, 41, 47, 53, 59, 71, 83, 89.$$

(Brute force in `reproduce.py` confirms each of these, and indeed every prime $p \leq 100$ with $p \equiv 2 \pmod{3}$.)

4 Cauchy–Davenport does not save the claim

The natural response is that an additive-combinatorial lower bound might force $3A$ to be everything. It does not. Cauchy–Davenport states that for $A, B \subseteq \mathbb{F}_p$,

$$|A + B| \geq \min(p, |A| + |B| - 1).$$

Applying it twice to $A + A + A$ gives

$$|3A| \geq \min(p, 3|A| - 2).$$

Proposition 8. *Let $p \equiv 2 \pmod{3}$ and $|A| = (p + 1)/3$. Then*

$$\min(p, 3|A| - 2) = p - 1 < p.$$

Proof. With $p = 3k + 2$ and $|A| = k + 1$ we get $3|A| - 2 = 3k + 1 = p - 1 < p$, so the minimum is $p - 1$. $\square$

Thus at the relevant size Cauchy–Davenport certifies only $|3A| \geq p - 1$; it leaves room for exactly one missing residue, and for the family of Theorem 5 the missing residue is $p - 1$. The bound is sharp but strictly weaker than full coverage, so it cannot rescue the conjecture. In particular the conjecture cannot be repaired by appealing to Cauchy–Davenport at the threshold.

5 Numerical table

p	$ A $	$(p - 1)/3$	$3A$	missing
5	2	4/3	$\{0, 1, 2, 3\}$	4
11	4	10/3	$\{0, 1, \dots, 9\}$	10
17	6	16/3	$\{0, 1, \dots, 15\}$	16

In each row $|A| > (p - 1)/3$ and yet $3A = \{0, 1, \dots, p - 2\}$ misses the nonzero residue $p - 1$, so the coverage claim fails.

6 Reproducibility

The brute-force checks are carried out by `reproduce.py`, which uses the Python 3 standard library only:

```
$ python3 reproduce.py
... (witness p = 5 , the family table, and the checks)
PASS: all checks verified; conjecture 00000002048 is FALSE.
```

It computes $3A$ for $A = \{0, 1\} \subset \mathbb{F}_5$ by enumerating all eight triples, asserts $4 \notin 3A$ and $|A| = 2 > 4/3$, and repeats the check for the general family $A = \{0, \dots, (p - 2)/3\}$ for every prime $p \leq 100$ with $p \equiv 2 \pmod{3}$, verifying in each case $3|A| > p - 1$, $3A = \{0, \dots, p - 2\}$ and $p - 1 \notin 3A$. It prints **PASS** and exits 0 exactly when every check holds.

7 Formalisation

The counterexample and the general family are formalised in the accompanying Lean 4 project under `lean4/`. It builds with `lake build` and is audited by `lake env lean Check.lean`, using core Lean only (no Mathlib) and no `sorry`. The 3-fold sum is defined concretely on residues of $\mathbb{F}_5$ represented as naturals reduced modulo 5, and the main formal statements are

```
theorem witness_sum3 : sum3 5 witnessA = [0,1,2,3]
theorem witness_misses_four : (sum3 5 witnessA).contains 4 = false
theorem conjecture_00000002048_false : coversNonzero 5 witnessA = false
theorem family_not_cover (p) (hp : 2 ≤ p) :
  coversNonzero p (familyA p) = false
theorem family_threshold (p) (hp : p % 3 = 2) :
  3 * (familyA p).length > p - 1
```

Here `witnessA` is $A = \{0, 1\}$ and `coversNonzero` is the conjecture’s coverage predicate. The general theorem `family_not_cover` shows that for every $p \geq 2$ the residue $p - 1$ is absent from the 3-fold sum of $\{0, \dots, (p-2)/3\}$, and `family_threshold` shows that this set satisfies the conjecture’s size hypothesis when $p \equiv 2 \pmod{3}$. The finite statements are discharged by `decide`; the general ones use the bound $x + y + z \leq 3(p-2)/3 \leq p-2 < p$ so that no reduction modulo p occurs. See `lean4/README.md` for the full statement table and scope.